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Prostate Imaging Standards for Screening Magnetic Resonance Imaging (PRISM)
International Consensus Recommendations

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IMPORTANCE Magnetic resonance imaging (MRI) is routinely used in prostate cancer diagnosis, but there is uncertainty about its optimal use in population-level screening.

OBJECTIVE To provide consensus recommendations on the acquisition, interpretation, and reporting of prostate MRI for cancer screening.

EVIDENCE REVIEW A systematic review and meta-analysis of randomized clinical trials and prospective cohort studies evaluating MRI for prostate cancer screening was conducted for these international consensus recommendations. A search of PubMed, CENTRAL, Scopus, Web of Science, and ClinicalTrials.gov, citation searching, and consultation with experts was performed in September 2024. Studies performing upfront MRI for the purpose of prostate cancer screening were included.

FINDINGS A total of 6 studies were identified and included 1919 participants, of whom 1426 underwent upfront screening MRI. Most studies used 3.0-T non–contrast-enhanced MRI. The pooled biopsy recommendation rate was 19.2% (95% CI, 11.7-26.7), with grade group (GG) 2 or higher prostate cancer detection of 6.0% (95% CI, 3.1-9.0) and GG1 cancer detection of 1.4% (95% CI, 0.7-2.2). The positive predictive value for GG2 or higher cancer was 36.3% (95% CI, 21.1-51.4). Findings informed a RAND/UCLA Appropriateness Method consensus exercise used to produce the Prostate Imaging Standards for Screening Magnetic Resonance Imaging (PRISM) recommendations. A total of 21 experts (8 urologists, 11 radiologists, and 2 pathologists from 6 countries) were included on the panel. Among 323 consensus statements, 235 (72.8%) reached agreement. If MRI is used in screening, it is recommended for men with an estimated life expectancy greater than 10 years and aged 50 to 70 years or from age 45 years in Black men. Screening MRI should be performed in men after a prostate-specific antigen (PSA) test, but there was no consensus on the optimal PSA threshold. Non–contrast-enhanced MRI with only T2-weighted and diffusion-weighted imaging was considered appropriate, with a maximum acceptable acquisition time of 15 minutes. The stage-gated (2-step) approach to reporting, where all MRI sequences are only revealed (second step) if a concordant focal lesion is identified on axial T2-weighted and high b-value diffusion imaging (first step), was recommended. Repeat screening should be risk stratified according to patient characteristics. Screening MRI should only be performed in accredited centers with radiologists meeting minimum reporting requirements and quality standards.

CONCLUSIONS AND RELEVANCE The PRISM consensus recommendations provide standardized expert guidance on the use of MRI in prostate cancer screening trials and future screening programs, including abbreviated non–contrast-enhanced MRI protocols, recommended screening intervals, stage-gated reporting, and quality assurance standards.

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Prostate cancer accounts for more than 1.4 million new diagnoses annually worldwide, with incidence projected to double by 2040.[1] Although screening with prostate-specific antigen (PSA) reduces mortality, it is associated with substantial harms, including unnecessary biopsies, overdiagnosis of indolent disease, and overtreatment.[2,3] As a result, most countries do not recommend population-level PSA screening and instead advocate shared decision-making.

Magnetic resonance imaging (MRI) has transformed the diagnostic pathway for suspected prostate cancer,[4] and randomized trials are now assessing its role as a primary screening test and as a reflex test following a PSA test.[5-7] However, there is a lack of guidance on the optimal implementation of MRI in screening. In the absence of definitive trial data, we convened an international multidisciplinary expert panel to develop consensus recommendations on the acquisition, interpretation, and reporting of screening prostate MRI to guide research and pilot prostate cancer screening initiatives.

Methods

We performed a consensus exercise using the RAND/UCLA Appropriateness Method (RAM). The methodology has been used widely in relation to the role of MRI in the diagnostic and treatment pathway for prostate cancer.[8-14] In the absence of clinical trial evidence, RAM produces clear recommendations for future validation. Our aim was to provide consensus on the acquisition, interpretation, and reporting of MRI performed for prostate cancer screening. The systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting guideline.

Systematic Review and Meta-Analysis

As per RAM guidance, a systematic review was conducted prior to the consensus exercise (PROSPERO identifier: CRD42024569877). The review aimed to assess the range of imaging protocols and reporting criteria used for screening MRI. It was restricted to studies using upfront MRI, as a systematic review by Fazekas et al[15] examining screening with MRI after PSA had been published recently. Our review, alongside that by Fazekas et al,[15] was used to inform statement development for the consensus exercise.

A search of PubMed, CENTRAL, Scopus, Web of Science, and ClinicalTrials.gov, citation searching, and consultation with experts was performed in September 2024. Studies performing upfront MRI for the purpose of prostate cancer screening were included. Two authors (N.M. and A. Light) screened abstracts and reviewed full-text articles, with disagreement settled by discussion. Only randomized clinical trials and prospective cohort studies were included. Risk of bias was assessed using the Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) tool.[16,17]

Meta-Analysis

We performed a meta-analysis assessing (1) biopsy recommendation; (2) adherence to biopsy recommendation; (3) International Society of Uropathology (ISUP) grade group (GG) 2 or higher prostate cancer detection rate (calculated as the number of GG2 or higher prostate cancer cases divided by the number undergoing MRI); (4) ISUP GG1 prostate cancer detection rate (calculated as the number of GG1 cases divided by the number undergoing MRI); (5) positive predictive value (PPV) of MRI for ISUP GG2 or higher detection (calculated as the number of GG2 or higher cases in men with positive MRI findings divided by the number with positive MRI findings); and (6) the ISUP GG2 or higher to GG1 detection ratio.

To account for the variation across studies, the primary biopsy recommendation threshold reported in each study was used (eTable 1 in Supplement 1). A random-effects meta-analysis pooled estimates across studies, reported with 95% CIs. The restricted maximum likelihood method was used to estimate between-study variance. Heterogeneity was assessed using Cochran Q and I^2 statistics with corresponding P values derived from the Cochran Q test. All statistical tests were 2-sided, and statistical significance was defined as $P < .05$. Sensitivity analyses evaluated differences in outcomes across thresholds (where reported), with leave-1-out meta-analyses to explore heterogeneity. Statistical analyses were performed using R version 4.4.0 (The R Foundation).

Consensus Exercise

The panel consisted of 8 urologists, 11 radiologists, and 2 pathologists from 6 countries; urologists and radiologists had extensive experience in MRI for prostate cancer screening or diagnosis (eTable 2 in Supplement 1).

For round 1, 305 statements were developed by a core group (N.M., A. Light, F.G., A.R.P., S.P., T.T.S., and H.U.A.). Participants were given 1 month to score statements independently. A set of assumptions was provided to each participant (eTable 3 in Supplement 1) with a summary of relevant evidence. Participants could provide suggestions for amending, adding, or deleting statements. Participants rated their agreement with statements on a 9-point Likert scale (1 indicating strong disagreement; 5 neither disagreement nor agreement; 9 strong agreement). Participants could abstain from statements outside their area of expertise. For each statement, agreement was calculated using the group median score (1-3 represented disagreement; 4-6, uncertainty; 7-9, agreement). Consensus was determined using the Interpercentile Range Adjusted for Symmetry method.

Round 2 was a hybrid consensus meeting in London, UK, on December 16, 2024, chaired by an independent clinical epidemiologist (J.v.d.M.). Statements were presented with a histogram of round 1 scores, including a report of the degree of agreement and consensus among participants, then discussed before rescoring. Modifications, additions, and deletions were permitted if approved by the chair. Scores from round 2 informed final consensus recommendations.

Results

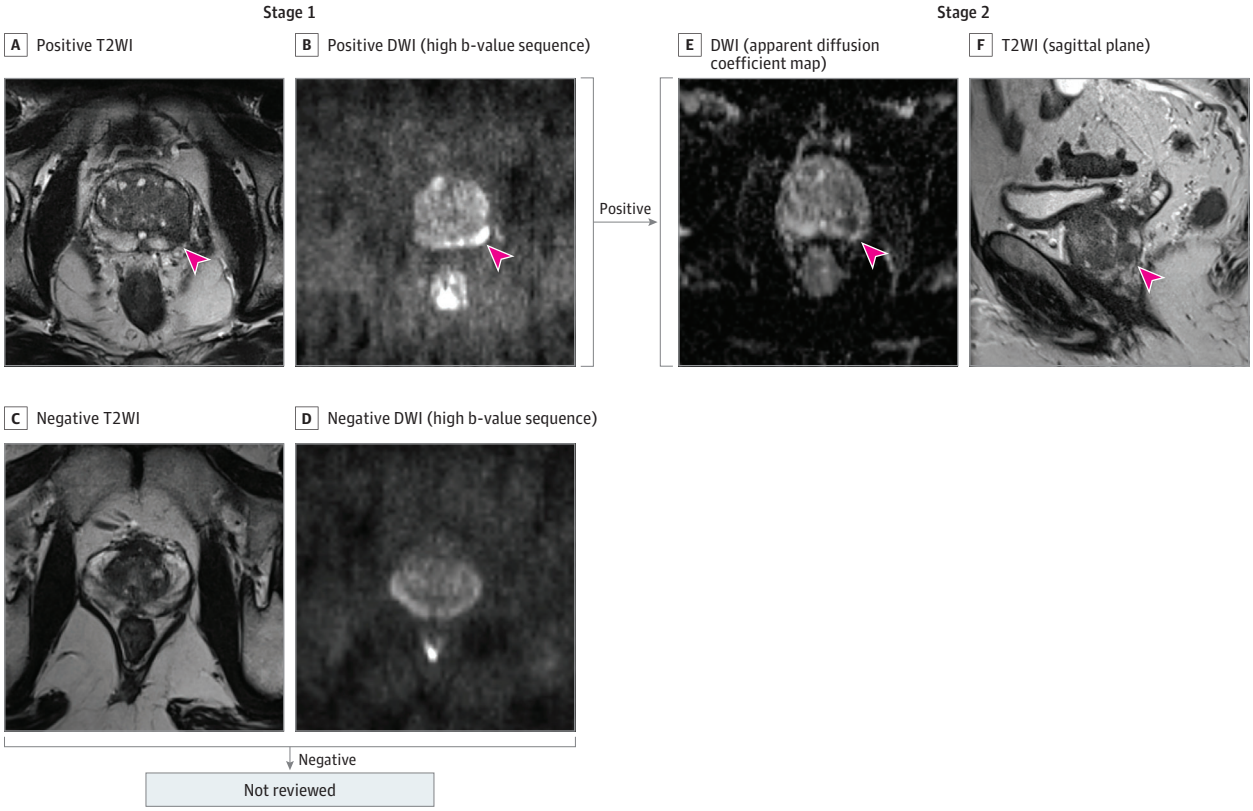
Systematic Review

The search yielded 2570 studies, of which 346 were duplicates. Of the remaining 2224 studies, 6 examined upfront screening MRI from 2016 to 2023 (2 randomized clinical trials and 4 cohort studies) (eFigure 1 in Supplement 1).[17-22] A total of 1919 participants were included, of whom 1426 (median [IQR] of 246 [175-303] per study) underwent MRI (eTables 4 and 5 in Supplement 1).

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Figure. Schematic of the Stage-Gated Approach to Interpretation of Screening Magnetic Resonance Imaging



In stage 1, only the T2-weighted imaging (T2WI; A and C) and high b-value diffusion-weighted imaging (DWI; B and D) sequences in the axial plane are reviewed. If a concordant lesion is identified (arrowheads), the full biparametric magnetic resonance imaging, including the remaining diffusion-weighted sequences (E) and planes (F), are reviewed and assigned a Prostate Imaging-Reporting and Data System score. If no lesion is seen in stage 1 (as in C and D), the scan does not move to stage 2 and is reported as negative.

Table 1. Summary of Outcomes and Estimates of Heterogeneity for Meta-Analysis at the Primary Threshold for Magnetic Resonance Imaging Positivity Used in Each Study

Outcome	Pooled outcome, % (95% CI)	Heterogeneity[a]	*P* value	I^2, % (95% CI)
Biopsy recommendation	19.2 (11.7-26.7)	Q_5 = 47.6	<.001	93.1 (80.8-93.7)
Biopsy adherence	84.7 (69.5-100)	Q_4 = 37.6	<.001	95.7 (70.4-98.7)
GG2 or higher detection	6.0 (3.1-9.0)	Q_5 = 23.4	<.001	83.4 (51.2-89.3)
GG1 detection	1.4 (0.7-2.2)	Q_5 = 6.5	.27	23.2 (0-74.8)
PPV (GG2 or higher)	36.3 (21.1-51.4)	Q_5 = 31.7	<.001	85.9 (67.2-91.4)
GG2 or higher to GG1 ratio	3.01 (1.90-4.76)	Q_5 = 5.3	.38	0 (0-71.7)

Abbreviations: GG, grade group; PPV, positive predictive value.

[a] Heterogeneity calculated using the Cochran *Q* statistic.

A 3.0-T MRI was used for most participants (1356 of 1426 [95%]), and 1376 of 1426 (96%) were non–contrast-enhanced MRIs. No study used endorectal coils. MRI protocols across studies are summarized in eTable 6 in Supplement 1. Acquisition time ranged from 10 to 30 minutes. A total of 4 studies used the Prostate Imaging-Reporting and Data System (PI-RADS) or modified PI-RADS score for reporting, 1 used the Likert scale and the modified PI-RADS score, and 1 used a novel stage-gated reporting system (**Figure**).[23,24] Risk of bias is summarized in eFigure 2 in Supplement 1.

Meta-Analysis

Forest plots are presented in eFigures 3 to 8 in Supplement 1, with pooled outcomes in **Table 1**. The MVP pilot study[17] offered biopsy to all patients regardless of MRI findings and so was excluded from the biopsy adherence meta-analysis.

Subgroup Analysis by Biopsy Recommendation Threshold

Stage-gated reading uses a stricter threshold for biopsy recommendation so was included in the PI-RADS 4 or greater subgroup for sensitivity analysis (eTables 7 and 8 in Supplement 1). The pooled biopsy recommendation rate for studies using the PI-RADS 3 or greater threshold was higher than those using PI-RADS 4 or greater (27.2% vs 16.0%). Adherence to biopsy was marginally lower (84.1% vs 89.2%). Studies using the PI-RADS 3 or greater threshold for biopsy detected GG2 or greater cancer and GG1 cancer in 4.5% and 1.8% of cases, respectively, compared with 7.0% and 1.1% for stud-

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ies using PI-RADS 4 or greater. PPV was lower for studies using the PI-RADS 3 or greater threshold (25.2% vs 40.3%).

Leave-1-Out Meta-Analysis

In the PPV meta-analysis, the ReIMAGINE study[19] contributed most to heterogeneity; excluding this study reduced the pooled PPV from 36.3% (95% CI, 21.1-51.4) to 26.2% (95% CI, 20.1-32.2), suggesting a significant role for stage-gated reporting used in this study.

Consensus Exercise

After round 1, 31 additional statements were included, 32 were updated, and 5 were removed for discussion and scoring in round 2. During the round 2 meeting, 30 statements were updated and 11 were removed, resulting in 323 statements scored (eTable in Supplement 2). In round 1, 200 of 305 statements (65.6%) reached consensus with either agreement or disagreement, increasing to 235 of 323 (72.8%) after round 2. These 235 statements were used to form the Prostate Imaging Standards for Screening Magnetic Resonance Imaging (PRISM) recommendations (**Box**).

PRISM Recommendations

Screening Population

Consensus was reached on the age at which screening should commence for the general population and Black men, but it was uncertain at what age screening should commence for those with a family history. While family history is a recognized risk factor, coding quality within electronic health records is of variable quality, making reliable population-level identification challenging.[25] It was uncertain if first screening MRI should be offered to men aged 71 to 75 years. Screening MRI is not recommended for men with a life expectancy of less than 10 years, reflecting the natural history of prostate cancer and improvements in advanced disease management. Although intuitive, life expectancy prediction may be difficult to operationalize, although the emergence of composite comorbidity indices that can be applied at scale to health records may provide a pragmatic solution.[26,27] Having lower urinary tract symptoms should not preclude a man from screening MRI, although participants acknowledged procedures for bladder outflow obstruction can impact interpretation of the peripheral zone on T2-weighted imaging (T2WI).

Pre-MRI Risk Stratification

Participants scored this section's statements primarily on clinical effectiveness and were permitted to consider or disregard cost. There was agreement that patients should undergo PSA before MRI. It is well-described that clinically significant prostate cancers can be detected below conventional PSA thresholds, but no consensus was reached on the most appropriate PSA threshold.[28]

MRI Technical Specifications

This section was scored assuming that screening MRI should not involve staging. Recommendations on MRI technical specifications were based on the principle that simple, short, and practical protocols are most appropriate for population-level screening. Patient preparation with enemas or antispasmodics is not compatible with large-scale population screening due to cost, workflow complexity, and possible adverse events. Similarly, endorectal coils should be avoided. Acquisition times should not exceed 15 minutes to main-

Box. Summary of Prostate Imaging Standards for Screening Magnetic Resonance Imaging (PRISM) Recommendations

Population

- Invite men aged 50-70 y to screening
- Invite Black men aged 45-49 y to screening
- Invite men with a family history or high-risk genetic predisposition to screening at an earlier age than the general population
- Do not invite men with an estimated life expectancy <10 y
- Do not invite men with negative prostate MRI or biopsy findings within 2 y
- Do not invite men with culture-proven urinary tract infection/prostatitis within 6 mo, but consider screening after resolution of infection
- The presence of lower urinary tract symptoms should not preclude men from screening

Pre-MRI Risk Stratification

- Use PSA to risk stratify men before screening MRI
- Do not routinely offer DRE prior to screening MRI
- Do not routinely offer biomarkers other than PSA before MRI
- Do not routinely offer polygenic risk scores before MRI

Technical Specifications

Patient preparation

- Do not routinely offer antispasmodics or enema prior to screening MRI
- Consider a low-fiber/low-residue diet 48 h prior to screening MRI

Technical requirements

- A 1.5-T field strength is adequate for screening MRI, provided scanners are configured to meet minimum technical standards
- A 3.0-T field strength is preferred
- An external (surface) phased array coil is required for screening MRI
- Do not use endorectal coils in screening MRI, regardless of the magnetic field strength

Protocol and sequences

- Screening MRI duration (acquisition time) should last no longer than 15 min
- Consider incorporating deep learning/artificial intelligence algorithms to accelerate sequence acquisition
- Do not use a dynamic contrast-enhanced sequence
- Omit the T1-weighted imaging sequence in screening MRI
- T2WI and DWI are mandatory sequences
- DWI should consist of an ADC map and high b-value imaging
- Use only 2 b values to calculate ADC to limit scan duration
- Use 1 low b-value set at 0-100 s/mm^2 (preferably 50-100 s/mm^2) and 1 intermediate b-value set at 800-1000 s/mm^2 (as per PI-RADS version 2.1) to calculate ADC
- The high b-value sequence should be set at ≥1400 s/mm^2
- Consider setting the high b-value sequence ≥1600 s/mm^2 on a 1.5-T scanner and ≥2000 s/mm^2 on a 3.0-T scanner
- Acquire the axial plane and 1 additional orthogonal plane (ie, sagittal/coronal)

(continued)

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Box. (continued)

Sequences should be acquired in the same imaging planes (axial and 1 other), with consistent orientation, angulation, and anatomical coverage

Unless otherwise stated, acquisition technique, slice thickness, and field of view should match or exceed the PI-RADS version 2.1 guidance

Interpretation

Screening MRI should only be interpreted by dedicated specialist readers with the following minimum level of experience:
- Diagnostic setting: 200 cases/y or 400 cases total
- Screening setting: 200 cases/y

Scans in men <50 y should be reviewed by expert readers

Reporting of all scans by 2 readers is not mandatory

If double reported, the second reader should be masked to the report of the first reader

If double reported and there is disagreement, a third highly experienced reader should arbitrate

After a positive screening MRI result, do not offer multiparametric MRI prior to biopsy

Positive screening scan results do not need routine multidisciplinary team review prior to biopsy

Readers should be provided the participant's age, PSA level, and results from previous rounds of screening

Use the stage-gated approach to report screening MRI

Assess only the axial T2WI and high b-value images in stage 1 of the stage-gated approach

Only move to stage 2 of the stage-gated approach if a focal lesion is identified on high b-value images and a corresponding focal lesion/diffuse region of interest is identified on T2WI

Assign a modified PI-RADS score to lesions in stage 2 of the stage-gated approach

Do not use the Likert scale to report screening MRI

Use a dedicated reporting template for screening MRI, including a pictorial representation of lesion size, location, and level of suspicion

Biopsy Considerations

Individuals with negative screen findings should not be offered biopsy and instead return to the screening program regardless of PSA density or other clinical risk factors

Recommend biopsy for those with a modified PI-RADS score of 4 or 5 in stage 2 of the stage-gated approach

Consider biopsy for those with a modified PI-RADS score of 3 in stage 2 of the stage-gated approach with elevated PSA density

Do not routinely use biomarkers other than PSA density to influence biopsy decision after screening MRI

Individuals with positive screen findings are not routinely required to undergo further review (either centrally or in secondary care) prior to biopsy recommendation

Individuals with negative screen findings do not require telephone or face-to-face consultation for results

Quality Control

All scans should undergo diagnostic image quality assessment using the criteria outlined in the stage-gated approach

(continued)

Box. (continued)

For nondiagnostic scans, repeated screening MRI or repeated sequences should be offered using the same sequences (eg, TSE for T2W, EPI for DWI)

In cases of persistent nondiagnostic image quality, alternative imaging techniques (eg, PROPELLER/BLADE for T2W, advanced motion-correction techniques for DWI) should be considered at the time of the initial scan

A maximum of 2 attempts at screening MRI should be offered before escalating to alternative imaging strategies or biopsy

Reporters should not provide a cancer likelihood score in nondiagnostic scans

Quality Assurance

Radiologists and radiographers/radiology technologists should participate in training programs that include prostate MRI image quality scoring

Screening MRI should only be performed at designated accredited centers

All readers, even those considered experts in the detection setting, must undergo certification to report screening MRI

A formal audit of MRI reading data should be undertaken annually

Phantom testing of MRI scanners used in screening should occur every 6 mo or annually

The DWI technical failure rate should be monitored, and cases of radiology-pathology discrepancies should be reviewed

Recommendation of reader and scanning center outcomes that should be audited periodically in a screening program incorporating MRI using at minimum the following variables:
- Number of screening MRI scans reported
- Number/proportion of indeterminate cases reported
- Ratio of positive to negative screening results
- Biopsy recommendation rate
- Ratio of grade group 2 or higher detection to grade group 1 detection
- Positive predictive value
- False-positive rates
- Proportion of nondiagnostic scans

Artificial Intelligence

Artificial intelligence models should be developed specifically for the screening population, considering the lower prevalence of high-grade cancers and the higher prevalence of indeterminate findings

The decision to biopsy should not be based solely on the interpretation of a screening MRI by an artificial intelligence tool

Patients should be informed when artificial intelligence contributes to their MRI interpretation

Abbreviations: ADC, apparent diffusion coefficient; DRE, digital rectal examination; DWI, diffusion-weighted imaging; EPI, echo-planar imaging; MRI, magnetic resonance imaging; PI-RADS, Prostate Imaging-Reporting and Data System; PROPELLER, periodically rotated overlapping parallel lines with enhanced reconstruction; PSA, prostate-specific antigen; T2WI, T2-weighted imaging; TSE, turbo spin echo.

tain scanner throughput, and 1.5-T or 3.0-T MRI can be used, provided it is optimized to minimum technical standards. T1-weighted sequences provide limited additional diagnostic information in biopsy-naive patients and can be foregone. Dynamic contrast-enhanced (DCE) sequences should be omitted as it is not practical

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Table 2. Recommendations for Risk-Adapted Rescreening Intervals

Scenario	Repeat screening interval, y
Negative MRI findings and no risk factors (low PSA density, low-risk racial and ethnic background, no family history)	4-5
Negative MRI findings, high-risk racial and ethnic background (eg, West African), and normal PSA density	3-4
Negative MRI findings and high PSA density	2-3
Negative MRI findings and high-risk genetic variant (eg, *BRCA1*/*BRCA2*)	2
Positive screening MRI finding, negative biopsy finding, and any PSA density	1

Abbreviations: MRI, magnetic resonance imaging; PSA, prostate-specific antigen.

to administer contrast in this setting. T2WI and diffusion-weighted imaging (DWI), including the apparent diffusion coefficient map and high b-value sequences, are the core sequences. It was acknowledged that the T2WI in-plane resolution recommended by PI-RADS version 2.1 may be overly restrictive in the screening setting.[23] Strict constraints on in-plane resolution can decrease the signal to noise ratio, making some scans more challenging to interpret. No consensus was reached regarding the exact standards to recommend for in-plane resolution, with some participants in favor of less restrictive recommendations and others concerned that less restriction would lead to smaller lesions being missed. Additional minor changes to the PI-RADS version 2.1 acquisition technique standards were suggested, but these fell outside the scope of the exercise. In addition to the axial plane, a minimum of 1 additional orthogonal plane (ie, sagittal and/or coronal) is mandatory, as this aids scan interpretation, particularly for lesions at the prostatic base and apex, facilitates cognitive prostate biopsy, and supports treatment planning.

MRI Interpretation

Participants agreed screening MRI should only be interpreted by specialist readers with a minimum level of experience, in line with European Society of Urogenital Radiology/EAU Section of Urologic Imaging consensus recommendations.[29] In contrast to mammography in breast cancer screening, MRI interpretation is more objective; therefore, double reporting is not recommended routinely. After positive screen MRI findings, performing multiparametric MRI prior to biopsy is unnecessary. While it was argued that multiparametric MRI may provide further information useful for treatment planning, it is overly burdensome as a primary screening test. In men younger than 50 years, screening MRI should only be interpreted by expert readers, as interpretation is complex and more likely to yield indeterminate scores with conventional scoring systems.

We recommend adoption of the stage-gated approach to reporting screening MRI. In stage 1, assessment is based on just axial T2WI and high b-value DWI. Stage 2 (remaining sequences and planes) is only revealed if concordant focal abnormalities are identified in stage 1, and a PI-RADS score is reported (eTable 9 in Supplement 1). This approach reduces indeterminate findings and biopsy recommendation rates and increases PPV.[30] While a small loss in sensitivity for lesions evident on apparent diffusion coefficient alone is possible, such tumors are likely to be small and identified at repeat screening.

Repeat Screening

This section was characterized by significant uncertainty, although participants agreed that more than 1 round of screening is mandatory. Intervals should be risk adapted. Agreement was not met for the exact screening intervals for each scenario presented to participants, but guidance for several scenarios is presented (**Table 2**).

Biopsy Considerations

Biopsy is recommended for those with PI-RADS scores of 4 to 5. Further risk stratification is recommended for indeterminate scans (eg, with PSA density or clinical risk factors), but there was uncertainty regarding the optimal PSA density cutoff. Blood-based or urine-based biomarkers should not be routinely used to influence biopsy decision after screening MRI.

Quality Assurance and Control

Individual and service quality data should be monitored by completing an annual audit of MRI reading data. This should be compared with peers, and if outcomes suggest retraining is required, readers may be asked to recertify.

Discussion

We report an international expert consensus on the use of MRI for prostate cancer screening. In the absence of trial data, the PRISM recommendations provide methodologically rigorous guidance on how MRI should be acquired, interpreted, and reported in screening trials and pilot programs.

In accordance with recommendations from the PRAISE-U initiative,[31] we recommend that any screening program for prostate cancer should incorporate prostate MRI and commence between ages 50 and 70 years, the upper bound of which is consistent with the core age group applied in the European Randomized Study of Screening for Prostate Cancer.[32] Long-term follow-up from European Randomized Study of Screening for Prostate Cancer suggests that commencing screening after age 70 years does not confer additional mortality benefit.[33,34] However, these observations are based on a trial that recruited more than 3 decades ago. Improvements in health-adjusted life expectancy may justify screening for selected older men.

We recommend that screening MRI should be used following a PSA test. The evidence for upfront MRI remains limited and heterogeneous, although a recent randomized trial reports that upfront MRI more than doubles the rate GG2 or higher cancer detection with favorable cost-effectiveness.[5] Long-term cancer-specific outcomes are still required to ascertain the effectiveness of upfront MRI. We did not reach consensus on the optimal PSA threshold for risk stratification. The addition of MRI and targeted biopsy to the screening pathway is likely to shift the balance in favor of applying a PSA value lower than 3.0 ng/mL (to convert to µg/L, multiply by 1). The Göteborg-2 Prostate Cancer Screening Trial, where MRI-targeted biopsy was performed for men with PSA values of 1.8 to 3.0 ng/mL, detected GG2 or higher disease in 4.6% of men without a measurable increase in overdiagnosis of GG1 disease compared with those with a PSA of 3.0 to 10.0 ng/mL.[35] Modeling from studies using upfront MRI suggests that a PSA threshold as low as 1.0 ng/mL fol-

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lowed by targeted-only biopsy may optimize this trade-off even further.[36] The TRANSFORM trial is directly comparing different pre-MRI PSA cutoffs, as well as upfront MRI in the absence of PSA.[37]

We also provide recommendations on rescreening intervals. Early PSA screening studies implemented fixed schedules for rescreening, but contemporary trials are increasingly adopting risk-adapted strategies tied to the most recent PSA value, ranging from 2 years to up to 8 years for men with very low baseline PSA and negative findings on work-up.[6,38-40] We recommend risk-adapted rescreening, including MRI findings, race and ethnicity, and germline genetic predisposition.

The stage-gated approach will be a new concept for many readers. Initially developed for use with 2 sequential MRI scans[30] (limited biparametric MRI followed by multiparametric MRI) then adapted for a single biparametric MRI, stage-gated reading is critical in reducing unnecessary biopsy by raising the bar for a positive result. On a cohort screened with upfront MRI, the stage-gated approach substantially improved PPV compared with reporting with PI-RADS (53% vs 30%) and halved the biopsy rate while maintaining cancer detection.[41] The algorithm requires prospective validation in a sequential screening pathway incorporating PSA. We recommend biopsy is offered to those with a PI-RADS score of 4 or 5 and omitted for a score of 3 in the absence of a high-risk PSA density. We were unable to recommend PSA density risk thresholds, as those used in the detection setting cannot be extrapolated to a screening cohort. Whether biopsy can be foregone for all PI-RADS score 3 cases remains unclear. Modeling from the IP1-PROSTAGRAM study suggests that omitting biopsy for all PI-RADS score 3 cases in those with a PSA of 1.0 ng/mL or greater would reduce biopsy rates from 115 to 71 per 1000 screened at the expense of 2 GG2 or higher cancers per 1000 screened and reduce GG1 detection from 7 to 2 per 1000 screened. We hypothesize that most missed cancers are likely to be small-volume GG2 tumors usually safe for active surveillance and likely to be detected in subsequent screening rounds, providing uptake and adherence to screening is high. Ongoing screening trials will address this evidence gap.

The successful implementation of MRI screening will depend not only on abbreviated, pragmatic protocols but also on rigorous quality assurance/quality control systems to ensure reproducibility across centers. Interpretation of screening MRI differs markedly from the diagnostic setting and is particularly challenging in younger men, where diffuse peripheral zone changes commonly return indeterminate clinical suspicion scores.[42] Image quality, particularly for DWI, is therefore critical. To reduce interreader variability, stringent reader training and use of accredited screening MRI centers, akin to the structured frameworks in breast cancer screening, are paramount.

Limitations

This study has several limitations. First, while we provide guidance on MRI acquisition, interpretation, and reporting in screening, the real-world implications of MRI-based screening must be considered. National implementation would require significant infrastructure investment with phased implementation incorporating pilot centers and account for variation in funding structures and demographic compositions across health care systems. Second, our recommendations define minimum standards for effective and safe screening with MRI for appropriately resourced settings and may not be feasible in resource-constrained environments. Further, our panel consisted predominantly of clinicians from academic institutions; implementation of screening with MRI should incorporate broad stakeholder representation, including experts in health equity and behavioral sciences, and clinicians from both academic and nonacademic centers. Third, although we recommend earlier and more frequent screening for Black men, this high-risk subgroup has been consistently underrepresented in major screening trials. Consequently, evidence guiding optimal screening strategies in Black men is insufficient. Addressing this gap is essential to ensure that screening strategies do not inadvertently widen disparities. Future trials must prioritize adequate representation of Black men to enable robust, subgroup-specific evaluation of benefits and harms. Fourth, we acknowledge that our recommendations are not formal clinical guidelines but expert opinion developed through robust methodology in the absence of definitive evidence. The PRISM recommendations must therefore be validated prospectively for whole-population screening and for high-risk subgroups.

Conclusions

The PRISM consensus recommendations provide standardized expert guidance on the use of MRI in prostate cancer screening trials and future screening programs, including abbreviated non–contrast-enhanced MRI protocols, recommended screening intervals, stage-gated reporting, and quality assurance standards.

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